

Microeconomics I

Class 2 — Monotonicity, convexity, indifference curves

Pavel Kocourek

Universitat de Barcelona · Master in Economics

30 September 2026

slide 1 of 33

Slide 1 Title

Preview version. The full slides and handout are posted after class. Class 2 covers one section of the textbook, MWG 3.B: the assumptions economists place on preferences over bundles of goods, and what each assumption does to the picture of indifference curves.

Plan for today

10:00	Sign-up sheet; three snacks
10:10	Warm-up: two relations
10:20	Quiz results: $\geq$ vs $\gg$, $\forall$ and $\exists$
10:30	Monotonicity, local non-satiation
11:00	Break
11:10	Indifference and contour sets
11:20	Convexity, the MRS
11:35	Five preferences, in pairs
11:50	Homothetic, quasilinear; summary

Slide 2 Plan for today

The morning at a glance. The first half sharpens tools: relations, vector inequalities, quantifiers, monotonicity. The second half draws the pictures: contour sets, convexity, five worked preferences. Everything today is MWG section 3.B; Muñoz-Garcia chapter 1 covers the same ground more gently.

Practicalities

- Group sign-up sheet going round: 2–5 names per group
- Problem set 1 out Tue 6 Oct, due in class Wed 14 Oct
- Office hours: today 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Pavel Kocourek

Microeconomics I · Class 2

3

slide 3 of 33

Slide 3 Practicalities

Problem sets are done in groups of 2 to 5; one line per group on the sign-up sheet.

Problem set 1 is handed out on Tuesday 6 October and is due in class on Wednesday 14 October, one paper copy per group.

Office hours: after class, up to 20 minutes per student; booking by e-mail gives priority.

Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

Three classes, three snacks

Wed 7 Oct
?

Wed 14 Oct
?

Wed 21 Oct
?

menu: chocolate bar · crisps · nuts · biscuits · fruit · wafer

Choose all three now. Hands up: three different ones?

Set-up after a published experiment; source given in class.

Pavel Kocourek

Microeconomics I · Class 2

4

slide 4 of 33

Slide 4 Three classes, three snacks

A thought experiment to open. You must order a snack for each of the next three Wednesday classes, all three now, from a menu of six. Would you pick three different ones?

Decide before class and keep your answer in mind. On Wednesday you see what people did in a real version of this experiment; slide 26 gives the economics.

Three snacks: what people chose

Revealed in class

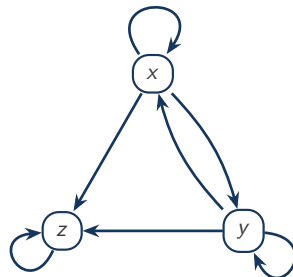
Before Wednesday: choose your three, and ask yourself whether you would choose differently if asked fresh each week.

Slide 5 Three snacks: what people chose

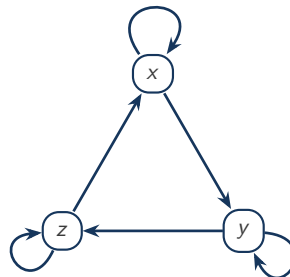
Revealed in class.

Beforehand: would you choose the same three if asked fresh each Wednesday, just before eating? Which of the two ways of choosing tells us more about what you like?

In pairs: two relations on $\{x, y, z\}$



R_1



R_2

$a \rightarrow b$ means $a R b$. Complete, reflexive, transitive, symmetric, asymmetric, antisymmetric?

Four minutes. Which one is a **rational** preference?

Slide 6 In pairs: two relations

Two relations on three objects, drawn as graphs: an arrow from a to b means $a R b$, and every node has a loop ($a R a$).

Try it: for each relation decide the six properties from class 1 (complete, reflexive, transitive, symmetric, asymmetric, antisymmetric), then whether it is a rational preference, that is complete and transitive.

It helps to list the pairs first. R_2 is $x R y, y R z, z R x$ plus the three loops.

Answers: two relations

Revealed in class

Try it in pairs before Wednesday: the six properties for R_1 and for R_2 , and which of the two is a rational preference.

Slide 7 Answers: two relations

Revealed in class.

Before Wednesday: list the pairs of each relation, check the six properties one by one, then decide which relation is complete and transitive. Keep your counterexamples; they are what we compare in class.

From class 1: what today builds on

$\succsim$ is **rational**: complete and transitive

MWG Def. 1.B.1

$x \succ y \iff x \succsim y \text{ and not } y \succsim x$ $x \sim y \iff x \succsim y \text{ and } y \succsim x$

$\sim$ is an equivalence relation: X splits into **indifference classes**

Prop. 1.B.1

Today: what those classes look like once $\succsim$ is **monotone** and **convex**.

Slide 8 From class 1

The three facts from class 1 that today rests on. A preference relation $\succsim$ is rational if it is complete (any two bundles can be compared) and transitive (no cycles). Strict preference $x \succ y$ means x is at least as good as y but not the other way round; indifference $x \sim y$ means each is at least as good as the other.

Because $\sim$ is an equivalence relation, the consumption set X splits into indifference classes.

Today's question: what shape do those classes have once we add monotonicity and convexity? With two goods they become the familiar indifference curves.

Diagnostic quiz: three things to fix today

Revealed in class

The results, and the three pieces of notation the quiz showed we need to fix.
Before Wednesday: look again at the quiz questions you were unsure about.

Slide 9 Diagnostic quiz

Revealed in class.

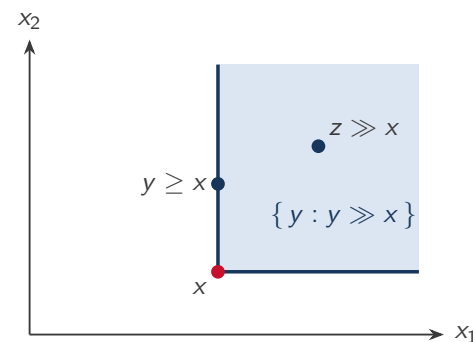
The diagnostic quiz is discussed on Wednesday. Beforehand, look again at the questions you were unsure about; the two slides that follow fix the notation they used.

Two ways to say “more”

$$y \geq x \iff y_1 \geq x_1 \text{ and } y_2 \geq x_2$$

$$y \gg x \iff y_1 > x_1 \text{ and } y_2 > x_2$$

Quiz: $x = (3, 1)$, $y = (3, 2)$.



Shaded: strictly more of both. Its edges belong to $\geq$ but not to $\gg$.

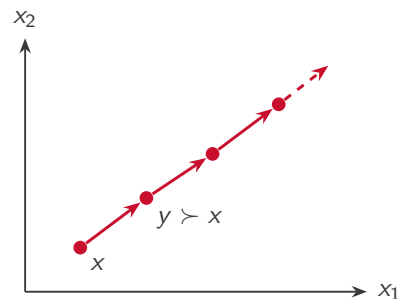
Slide 10 Two ways to say “more”

Two inequalities between bundles. $y \geq x$ means y has at least as much of every good as x ; a tie in some good is allowed. $y \gg x$ means y has strictly more of every good.

In the picture the shaded open quadrant is the set of bundles $\gg x$. Its two edges (same amount of one good, more of the other) satisfy $\geq$ but not $\gg$. Quiz example: $y = (3, 2)$ versus $x = (3, 1)$: $y \geq x$ holds, $y \gg x$ fails because good 1 ties.

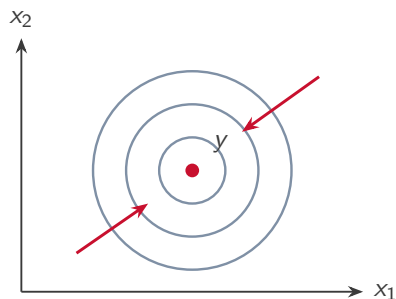
The combination “ $y \geq x$ and $y \neq x$ ” (at least as much of both, strictly more of at least one) is exactly what strong monotonicity uses on slide 13.

Swap the quantifiers, change the claim



$$\forall x \exists y : y \succ x$$

for every bundle, a better one



$$\exists y \forall x \neq y : y \succ x$$

one bundle beats all: a bliss point

Left: no bundle is best — **non-satiation**. Its local version arrives in ten minutes.

Slide 11 Swap the quantifiers

Same symbols, opposite claims. Left: for every bundle x there is a y with $y \succ x$. No bundle is best; wherever you are, something beats it (the chain of arrows). Right: there is one y that beats every other bundle, a bliss point; the circles around it are its indifference curves.

Swapping $\forall$ and $\exists$ is the whole difference between “no best bundle” and “a best bundle”.

The left statement is non-satiation. Its local version, slide 15, reads “for all x , for all ε , there exists y ”.

Slide 12 Monotonicity

Part 1: three versions of “more is better” (MWG 3.B), and the weakest one that still gives thin indifference curves.

Monotonicity

MWG 3.B

11
slide 11 of 33

slide 12 of 33

Definition: monotone, strongly monotone

$\succsim$ on $X = \mathbb{R}_+^2$ is **monotone** if

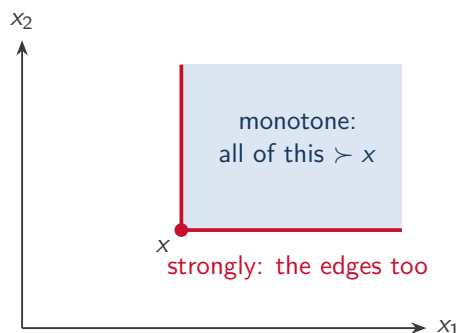
$$y \gg x \Rightarrow y \succ x$$

strongly monotone if

$$y \geq x \text{ and } y \neq x \Rightarrow y \succ x$$

MWG Definition 3.B.2.

Strongly monotone $\Rightarrow$ monotone. The converse fails — an example in the exercise.



Slide 13 Definition: monotone, strongly monotone

MWG Definition 3.B.2, on the consumption set $X = \mathbb{R}_+^2$ (two goods, non-negative amounts). $\succsim$ is monotone if a bundle with strictly more of every good is strictly preferred: $y \gg x$ implies $y \succ x$. It is strongly monotone if a bundle with at least as much of every good and more of at least one is already strictly preferred: $y \geq x$ and $y \neq x$ imply $y \succ x$.

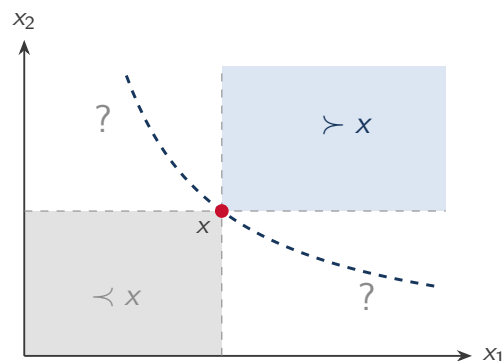
Picture: monotonicity claims the open shaded quadrant above and to the right of x ; strong monotonicity also claims its two edges.

Strongly monotone implies monotone, because $y \gg x$ is a special case of $y \geq x$, $y \neq x$. The converse fails: one of the five preferences of Exercise 1 (slide 18) shows it.

Slide 14 Where can the indifferent bundles be?

What monotonicity does to the indifference curve through x . Everything strictly north-east of x is strictly better (blue). Everything strictly south-west is strictly worse (grey), because there $x \gg y$. So a bundle indifferent to x can only lie in the two remaining quadrants or on their edges: the curve through x cannot slope upward. Monotonicity alone still allows flat or vertical stretches along the edges. Strong monotonicity removes them, and the curve is strictly downward sloping. Which of the five preferences on slide 18 has such stretches?

Monotone: where can the indifferent bundles be?



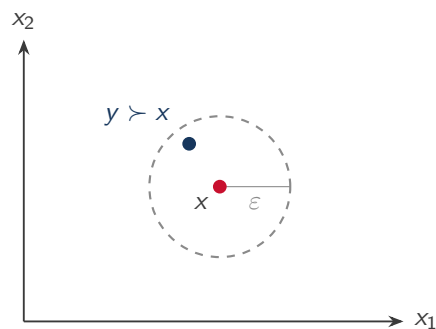
Nothing indifferent to x lies strictly north-east or strictly south-west of it. The curve through x cannot slope up.

Definition: local non-satiation

$\succsim$ is **locally non-satiated** if
for every $x \in X$ and every $\varepsilon > 0$
there is $y \in X$ with
 $\|y - x\| \leq \varepsilon$ and $y \succ x$

MWG Definition 3.B.3.

$\forall x \forall \varepsilon \exists y$: the quiz's quantifier order. No direction is prescribed: y may have **less** of a good.



Slide 15 Definition: local non-satiation

MWG Definition 3.B.3. $\succsim$ is locally non-satiated if for every bundle x and every distance $\varepsilon > 0$, however small, there is a bundle y within distance ε of x that is strictly better than x . Read the quantifiers in order: for all x , for all ε , there exists y .

No direction is prescribed. In the picture y has less of good 1 and more of good 2. So local non-satiation allows “bads” such as garbage or pollution, as long as something nearby is better.

Where does it fail for the bliss-point preference of slide 11? Think about it before Wednesday; the definition says “for every x ”.

Slide 16 Three strengths of “more is better”

Strongly monotone implies monotone, which implies locally non-satiated; neither arrow reverses. The grey question marks stand for the counterexamples, revealed in class: preferences of Exercise 1 (slide 18) that have the weaker property but not the stronger one. Which are they?

Also try: prove that monotone implies locally non-satiated in two lines, and find the step where $X = \mathbb{R}_+^2$ is used. Discussed in class.

Three strengths of “more is better”

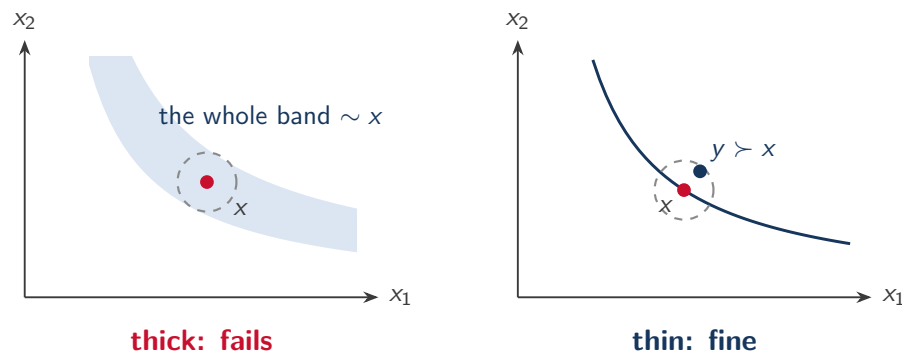


Prove monotone $\Rightarrow$ locally non-satiated in two lines. Where do you use $X = \mathbb{R}_+^2$?

Grey: a preference with the weaker property but not the stronger one. Which of the exercise's five are they?

Revealed in class.

Local non-satiation rules out thick indifference sets



Inside a thick band, every bundle within ε of x is indifferent to x .

After MWG Figure 3.B.2, redrawn; the circle test on the right is MWG Figure 3.B.1.

Slide 17 Thick indifference sets

Why the book bothers with local non-satiation. Left: a thick band of mutually indifferent bundles. From a bundle x inside the band, every bundle within a small radius is indifferent to x , so no strictly better bundle exists nearby: local non-satiation fails. Right: a thin curve; any circle around x reaches the better side.

Local non-satiation is the weakest of the three assumptions that rules out thick indifference sets, and it is all that Walras' law will need later in the course. After MWG Figure 3.B.2; the circle test is MWG Figure 3.B.1.

In pairs: monotone? strongly? locally non-satiated?

$x \succsim y \iff$	monotone	strongly monotone	LNS
$x_1 + x_2 \geq y_1 + y_2$			
$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$			
$x_1 - x_2 \geq y_1 - y_2$			
$\ x - (5, 5)\ \leq \ y - (5, 5)\ $			
$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$			

Five minutes. Give a counterexample for every $\times$.

Answers revealed in class. Try it in pairs before Wednesday.

Slide 18 In pairs: monotone? strongly? LNS?

Five preferences, each given by the rule for " x is at least as good as y ". Fill the row for each before Wednesday and find a counterexample for every cross you put; one bundle where the property fails is enough.

Revealed in class: the ticks and the counterexamples. A question to take with you: do any of the five have thick indifference sets?

Slide 19 Break

Back at 11:10.

Break

Back at 11:10

slide 19 of 33

Slide 20 Indifference and contour sets

Part 2: the sets an indifference curve is made of, and why curves do not cross. MWG 3.B.

Indifference and contour sets

MWG 3.B

slide 20 of 33

Three sets attached to a bundle x

$$\{y \in X : y \succeq x\}$$

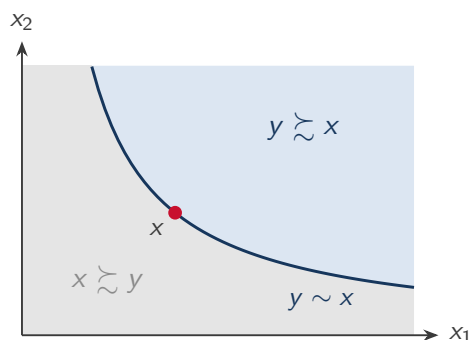
all bundles y in X such that y is at least as good as x

upper contour set $\{y \in X : y \succeq x\}$

indifference set $\{y \in X : y \sim x\}$

lower contour set $\{y \in X : x \succeq y\}$

MWG 3.B, p. 43, in the text; Figure 3.B.2(b).



Slide 21 Three sets attached to a bundle

MWG 3.B, p. 43 (defined in the text, drawn in Figure 3.B.2(b)), and the notation from the quiz. Read $\{y \in X : y \succeq x\}$ as “the set of all bundles y in X such that y is at least as good as x ”: the braces build a set, the colon reads “such that”, the condition selects. The budget set $\{x \in \mathbb{R}_+^2 : p \cdot x \leq w\}$ uses the same grammar.

Three sets attached to a bundle x : the upper contour set, all bundles at least as good as x (blue, on and above the curve); the indifference set, all bundles indifferent to x (the curve); the lower contour set, all bundles that x is at least as good as (grey, on and below). Note that x belongs to all three and the curve belongs to both contour sets.

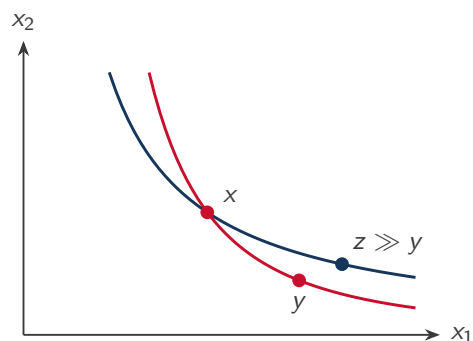
Which of the three is a budget set most like? Think about it; asked in class.

Slide 22 Indifference curves do not cross

Why two indifference curves cannot cross when $\succeq$ is rational and monotone. Suppose they cross at x . Pick y on the red curve and z on the blue curve with z strictly north-east of y , so $z \gg y$. Since $y \sim x$ and $x \sim z$, transitivity gives $y \sim z$. But $z \gg y$, so monotonicity gives $z \succ y$. Contradiction.

Each axiom does one job: transitivity forbids the crossing, monotonicity orders the curves. So exactly one indifference curve passes through each bundle, and curves further from the origin are better.

Indifference curves do not cross



$$y \sim x \text{ and } x \sim z \Rightarrow y \sim z$$

transitivity

$$z \gg y \Rightarrow z \succ y$$

monotone

Contradiction.

Rational + monotone: one indifference curve through every bundle; further out is better.

Convexity

MWG 3.B

slide 23 of 33

Slide 23 Convexity

Part 3: convexity, a taste for mixing, and the marginal rate of substitution. MWG 3.B.

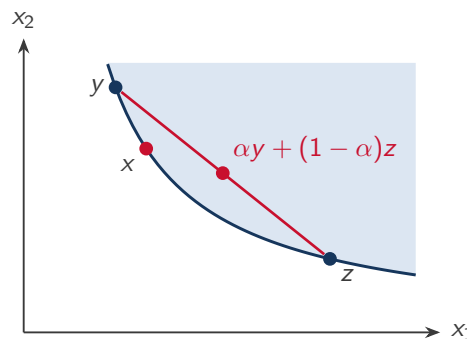
Definition: convex preferences

$\succsim$ is **convex** if every upper contour set $\{y \in X : y \succsim x\}$ is a convex set:

$$y \succsim x \text{ and } z \succsim x \\ \Rightarrow \alpha y + (1 - \alpha)z \succsim x, \quad \alpha \in [0, 1]$$

MWG Definition 3.B.4.

Averages are at least as good as extremes. The curve bows toward the origin.



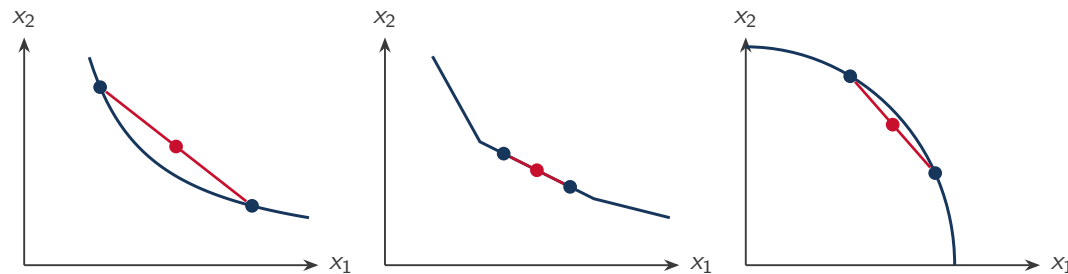
Slide 24 Definition: convex preferences

MWG Definition 3.B.4. $\succsim$ is convex if every upper contour set is a convex set: whenever y and z are both at least as good as x , so is every weighted average $\alpha y + (1 - \alpha)z$ with α between 0 and 1. In the picture y and z lie on the curve through x , and the red chord between them stays inside the blue set.

In words: averages are at least as good as extremes, and the indifference curve bows toward the origin. This is the convex set of class 1 (the segment between any two points of the set stays in the set), applied to the upper contour set of every bundle.

Is asking instead that the lower contour set be convex the same property? Think about it; asked in class, and the next slide holds a clue.

Strictly convex, convex, not convex



strictly convex

the mix is strictly better

convex, not strictly

a flat stretch: the mix is only $\sim$

not convex

the mix is worse

strictly convex: $y \succsim x, z \succsim x, y \neq z \Rightarrow \alpha y + (1 - \alpha)z \succ x, \alpha \in (0, 1)$

MWG Definition 3.B.5. Blue curve: an indifference set; blue dots: two bundles on it; red dot: their mix.

Slide 25 Strictly convex, convex, not convex

MWG Definition 3.B.5. $\succsim$ is strictly convex if, whenever y and z are two different bundles both at least as good as x , every mix $\alpha y + (1 - \alpha)z$ with $0 < \alpha < 1$ is strictly better than x .

Three panels, each with two blue bundles on an indifference curve and their mix in red. Strictly convex: the chord lies strictly above the curve, so the mix is strictly better. Convex but not strictly: along a linear stretch the chord stays on the curve, so the mix is only indifferent (which of the five preferences on slide 28 have such a stretch? Check the axes too). Not convex: the curve bows away from the origin, the chord dips below it, and the mix is worse than the endpoints.

Back to the snacks

Revealed in class

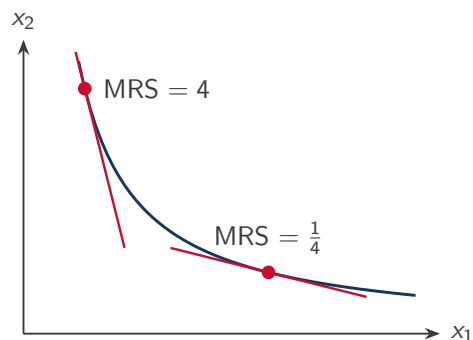
Think beforehand: if you are indifferent between two bundles and your preferences are convex, what does that say about a mix of the two?

Slide 26 Back to the snacks

Revealed in class.

Beforehand: could three weeks of snacks be treated as one bundle? If so, what does convexity say about mixing?

The marginal rate of substitution is the slope



$MRS_{12}(x)$: good 2 given up per unit of good 1, staying indifferent
 $= -\text{slope of the curve at } x$

Monotone + convex: moving right along a curve, the MRS **never rises**; strictly convex: it falls.

With a utility function u : $MRS_{12} = (\partial u / \partial x_1) / (\partial u / \partial x_2)$, the quiz's last question.

Slide 27 The MRS is the slope

The marginal rate of substitution of good 1 for good 2 at x , $MRS_{12}(x)$, is how many units of good 2 the consumer gives up for one more unit of good 1 while staying indifferent. Geometrically it is minus the slope of the indifference curve at x .

Picture: one indifference curve. At $(1, 4)$ the tangent has slope -4 , so the MRS is 4: good 1 is scarce and valued highly. At $(4, 1)$ the slope is $-\frac{1}{4}$, so the MRS is $\frac{1}{4}$.

With monotone and convex preferences (and smooth curves) the MRS never rises as you move right along a curve; with strictly convex preferences it falls. "Diminishing MRS" is convexity in other words. The grey line previews the formula from the quiz's last question, the ratio of marginal utilities; it arrives with utility functions on Tuesday.

In pairs: five preferences on $\mathbb{R}_+^2$

	$x \succsim y \iff$	monotone	strongly	convex	strictly
perfect substitutes	$x_1 + x_2 \geq y_1 + y_2$				
perfect complements	$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$				
Cobb–Douglas type	$x_1 x_2 \geq y_1 y_2$				
quasilinear	$x_1 + \sqrt{x_2} \geq y_1 + \sqrt{y_2}$				
lexicographic	$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$				

Sketch the indifference set through $(2, 2)$ for each, then fill the row. Eight minutes.

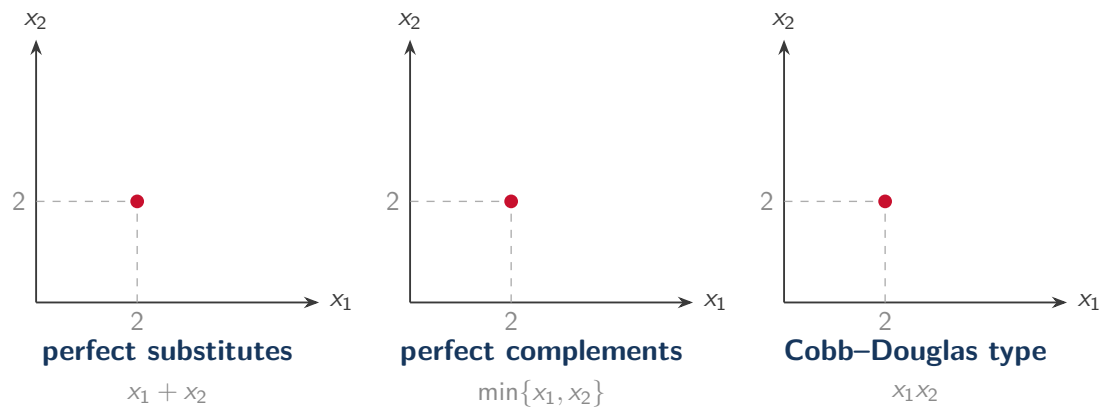
Slide 28 In pairs: five preferences

Five standard preferences on $\mathbb{R}_+^2$, axes included, each given by the rule for " x is at least as good as y ".

Try it: for each preference sketch the indifference set through $(2, 2)$, then decide whether it is monotone, strongly monotone, convex, strictly convex. The sketches matter more than the ticks: most properties can be read off the shape, and the crosses all need a counterexample.

Answers in class: the maps first, then the table with reasons. Bring your sketches.

Answers: the indifference maps, I



Revealed in class. Before Wednesday, sketch the indifference set through the red dot (2, 2) in each panel.

Pavel Kocourek

Microeconomics I · Class 2

29

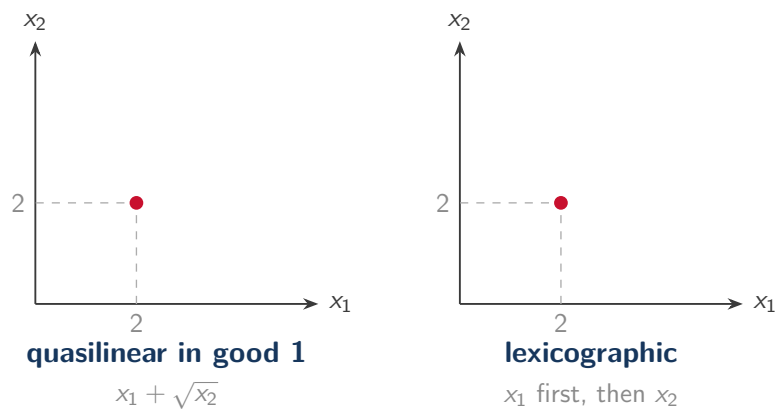
slide 29 of 33

Slide 29 Answers: the indifference maps, I

Revealed in class.

Before Wednesday, sketch the indifference set through (2, 2) for perfect substitutes, perfect complements and the Cobb–Douglas type, in the empty panels on the slide. Then add two more indifference sets of each, to see the whole map.

Answers: the indifference maps, II



Revealed in class. Same task: the indifference set through (2, 2). For the lexicographic one, also shade the upper contour set.

Pavel Kocourek

Microeconomics I · Class 2

30

slide 30 of 33

Slide 30 Answers: the indifference maps, II

Revealed in class.

Same task for the quasilinear and the lexicographic preference. For the lexicographic one, also shade the upper contour set of (2, 2).

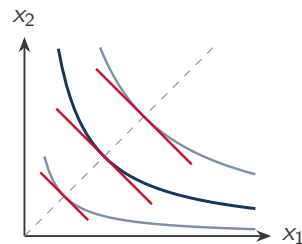
Revealed in class

Fill the table of slide 28 in pairs before Wednesday, with a counterexample for every cross you put.

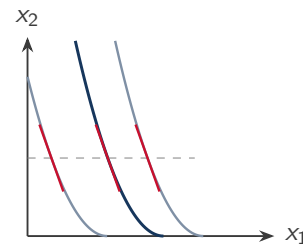
Revealed in class.

Fill the table on slide 28 in pairs before Wednesday, with a counterexample for every cross. Check each property on the whole of $\mathbb{R}_+^2$, axes included.

Two families you will meet again



homothetic
same slope along a ray



quasilinear in good 1
same slope along a horizontal

homothetic: $x \sim y \Rightarrow \alpha x \sim \alpha y, \alpha \geq 0$

quasilinear in good 1: $x \sim y \Rightarrow x + \alpha e_1 \sim y + \alpha e_1, \alpha \in \mathbb{R}; x + \alpha e_1 \succ x, \alpha > 0$

MWG Def. 3.B.6 (monotone $\succsim$ on $\mathbb{R}_+^2$), Def. 3.B.7 (on $X = \mathbb{R} \times \mathbb{R}_+$, so $\alpha < 0$ stays in X); $e_1 = (1, 0)$.

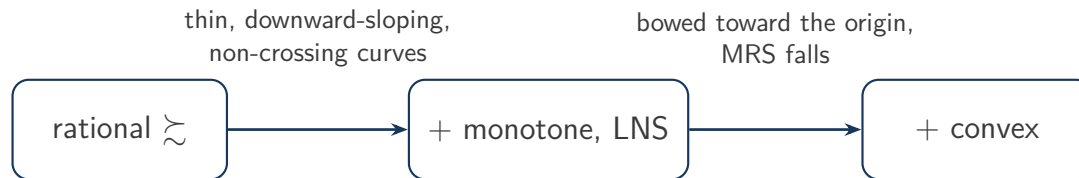
Slide 32 Two families you will meet again

MWG Definitions 3.B.6 and 3.B.7; only 3.B.6 assumes a monotone preference. Homothetic: if $x \sim y$ then $\alpha x \sim \alpha y$ for every $\alpha \geq 0$. The whole indifference map is one curve scaled along rays from the origin, so along any ray the slope (the MRS) is the same: the red tangents on the 45-degree line are parallel. Cobb–Douglas, perfect substitutes and perfect complements are all homothetic.

Quasilinear with respect to good 1: if $x \sim y$ then $x + \alpha e_1 \sim y + \alpha e_1$ for every α , where $e_1 = (1, 0)$, and adding good 1 always makes a bundle strictly better. The curves are horizontal translates of each other, so along any horizontal line the slope is the same. MWG states this definition on $X = \mathbb{R} \times \mathbb{R}_+$, so that shifting left by α keeps the bundle in X .

Both families return with demand, in the income expansion paths of weeks 2 and 3.

Today in one line



Tuesday: when a number $u(x)$ can stand in for $\succsim$, and why the lexicographic one has none.
Problem set 1 goes out.

Read MWG 3.C (or Muñoz-Garcia ch. 1) for Tuesday 6 October.

Slide 33 Today in one line

Start with a rational $\succsim$. Add monotonicity or local non-satiation, and the indifference sets are thin, downward-sloping, non-crossing curves ordered outward. Add convexity, and the curves bow toward the origin, with a falling MRS.

Tuesday 6 October: utility representation (MWG 3.C). When can a number $u(x)$ stand in for $\succsim$, why the lexicographic preference has no such number, continuity, quasiconcavity. Problem set 1 goes out on Tuesday.

Reading for Tuesday: MWG 3.C, or Muñoz-Garcia chapter 1.